

7.1.1

For two arbitrary intervals $X = (x_1, x_2), Y = (y_1, y_2)$ at least one of the following holds:

- X and Y are disjoint and $X \rightarrow Y$
- X and Y are disjoint and $Y \rightarrow X$
- X and Y overlap.

Proof: x_1, x_2, y_1, y_2 are events. Precedence is a *strict* total order on events, so for an arbitrary interval $I = (a, b)$ we have $a \rightarrow b$ (allowing $b \rightarrow a$ only makes the definition of an interval confusing and harder to reason about, so it's irrelevant).

If $x_2 \rightarrow y_1 \vee y_2 \rightarrow x_1$ is false, i.e. $y_1 \rightarrow x_2 \wedge x_1 \rightarrow y_2$, then X and Y overlap.

Let's examine the cases where X and Y are disjoint:

- $x_2 \rightarrow y_1$: by definition, $X \rightarrow Y$
- $y_2 \rightarrow x_1$: by definition, $Y \rightarrow X$.

□

7.1.2

Relation of precedence on intervals is

1. Irreflexive: $\forall X \neg(X \rightarrow X)$
2. Antisymmetric: $\forall X, Y \ X \rightarrow Y \Rightarrow \neg(Y \rightarrow X)$
3. Transitive: $\forall X, Y, Z \ (X \rightarrow Y \wedge Y \rightarrow Z) \Rightarrow X \rightarrow Z$

Proof:

1. Assume an arbitrary interval $X = (x_1, x_2)$. $X \rightarrow X$ means $x_2 \rightarrow x_1$, which contradicts with $x_1 \rightarrow x_2$ shown above, so $X \rightarrow X$ is impossible.
2. Assume $X = (x_1, x_2), Y = (y_1, y_2)$ and $X \rightarrow Y$. By definition of precedence on intervals we have $x_2 \rightarrow y_1$ and $y_2 \rightarrow x_1$. Earlier we have shown that $\forall I \ I = (a, b) \Rightarrow a \rightarrow b$, so $x_1 \rightarrow x_2$ and $y_1 \rightarrow y_2$. Combining these relations, we get

$$x_2 \rightarrow y_1 \rightarrow y_2 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots$$

which leads to contradiction with irreflexivity of precedence on events.

3. Assume $X = (x_1, x_2), Y = (y_1, y_2), Z = (z_1, z_2)$, $X \rightarrow Y$ and $Y \rightarrow Z$. By definition of precedence on intervals:

$$X \rightarrow Y \Leftrightarrow x_2 \rightarrow y_1 \quad Y \rightarrow Z \Leftrightarrow y_2 \rightarrow z_1$$

Combining these relations and the fact that $y_1 \rightarrow y_2$:

$$x_2 \rightarrow y_1 \rightarrow y_2 \rightarrow z_1 \Rightarrow x_2 \rightarrow z_1$$

□

Example of $X \neq Y$ where $\neg(X \rightarrow Y) \wedge \neg(Y \rightarrow X)$:

Let $X = (x_1, x_2), Y = (y_1, y_2)$, $x_1 \rightarrow y_1 \rightarrow x_2 \rightarrow y_2$. $x_1 = y_1$ and $x_2 = y_2$ are impossible due to irreflexivity, so $X \neq Y$. We have $x_1 \rightarrow y_2$, which disallows $y_2 \rightarrow x_1$ and $y_1 \rightarrow x_2$, which disallows $x_2 \rightarrow y_1$, leading to the conclusion that neither $X \rightarrow Y$ nor $Y \rightarrow X$ is true.